[TODO: Title]: Capability-Structured Agentic Multi-Agent Control of Conditional Handover and PDCP Duplication in LEO Networks

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I. Introduction

Low Earth orbit (LEO) satellite networks are emerging as a key component of non-terrestrial networks (NTNs) toward ubiquitous sixth-generation (6G) connectivity, owing to their wide-area coverage and substantially lower propagation delay than conventional geostationary satellite systems [1], [2]. Their rapid orbital motion, however, continuously changes satellite visibility and link quality, forcing user equipments (UEs) to repeatedly transfer their serving links across satellites. As LEO constellations become denser, a UE may simultaneously observe multiple candidate satellites, while many UEs compete for the finite radio and control-plane resources of the same satellites. Mobility management therefore becomes not merely a problem of selecting the strongest satellite, but a sequential decision problem that couples service continuity, resource contention, and handover overhead.

The Third Generation Partnership Project (3GPP) provides two mechanisms that are particularly relevant to this problem. Conditional handover (CHO) allows candidate configurations to be prepared before the actual handover execution, thereby separating handover preparation from the eventual serving-cell switch [24]. At the data plane, packet data convergence protocol (PDCP) duplication allows the same PDCP data to be transmitted over multiple lower-layer legs to improve delivery reliability [25]. These mechanisms provide useful degrees of freedom for highly mobile NTN environments, but they also create a nontrivial control problem. Preparing many candidate links reserves network resources, persistent duplication consumes additional radio capacity, whereas switching too aggressively increases handover and ping-pong events. The desirable configuration therefore depends jointly on the UE's current service condition, the anticipated lifetime of a candidate link, and the congestion created by other UEs.

A substantial body of work has addressed LEO mobility using measurement- or location-based rules, graph optimization, and learning methods [5]–[8]. More recent approaches exploit temporal satellite mobility or multi-connectivity to improve service continuity. For example, GEMS jointly models spatial UE–satellite association and temporal mobility evolution for maintaining dual connectivity [10], while learning-based schemes employ deep or multi-agent reinforcement learning to adapt satellite handover decisions to changing propagation conditions and network load [11], [12]. These studies establish the importance of mobility awareness and distributed decision making, but target selection and handover timing are typically treated as the principal control variables. Meanwhile, multi-connectivity and multipath schemes exploit simultaneous links for reliability and continuity [16]. What remains less explored is the *joint, stateful control of CHO preparation, PDCP duplication, and primary-link switching* when multiple UEs contend for shared satellite resources under delayed and imperfect telemetry.

This challenge is further complicated by partial observability. Instantaneous measurements alone do not reveal whether a degradation is transient or persistent, and a locally attractive handover target can become unavailable before the preparation and switching procedure completes. Moreover, independent UE decisions are coupled through shared satellite capacity: simultaneous preparations and duplicated transmissions toward the same satellite can make individually reasonable actions collectively inefficient. Consequently, effective mobility control requires not only reactive action selection but also temporal memory, anticipation of delayed consequences, and lightweight coordination among distributed decision makers.

These requirements are closely aligned with the emerging concept of Agentic AI. Recent wireless research characterizes agentic systems through autonomous perception, reasoning, goal-directed action, and coordination, with both large-language-model (LLM)-based and reinforcement-learning-based realizations [4], [18]. In satellite networks, AERS demonstrates an agentic multi-agent architecture for decentralized beam and resource scheduling [2], whereas LLM-based frameworks employ semantic reasoning for satellite resource orchestration and SAGIN management [3], [20]. Other recent studies combine LLMs with MAPPO for cognitive satellite–terrestrial resource allocation [21] or apply agentic decision making to multipath LEO transmission [22]. These works demonstrate the potential of Agentic AI in communication networks, but their primary focus is resource orchestration, intent interpretation, routing, or transport control rather than protocol-level LEO mobility.

Motivated by this gap, we develop a capability-structured agentic multi-agent framework for joint CHO and PDCP-duplication control in shared-capacity LEO networks. Each UE acts autonomously using only its locally available telemetry and protocol state, while centralized training with decentralized execution (CTDE) is used to learn cooperative policies. Rather than relying on a monolithic LLM, the framework explicitly instantiates the capabilities required by the mobility problem: recurrent memory summarizes temporally correlated local experience; a protocol-aware analytical reasoner exploits

deterministic satellite ephemeris to anticipate the delayed consequences of candidate actions; and lightweight congestion prices communicate shared-resource pressure without exposing the states or actions of other UEs. A training-time primal–dual controller further regulates the reliability objective. This decomposition yields the M1–M5 family and allows the roles of perception, memory, reasoning, coordination, and goal regulation to be examined separately.

The main contributions of this work are summarized as follows.

- We formulate joint CHO preparation, PDCP-duplication control, and primary-satellite switching in a shared-capacity LEO network as a constrained decentralized partially observable Markov decision process. The model explicitly captures preparation contention, duplication overhead, handover cost, delayed and noisy telemetry, persistent hidden link degradation, and stochastic traffic and capacity disturbances.
- We develop a capability-structured agentic control framework built on CTDE-MAPPO. Beyond a reactive local policy, the framework incorporates recurrent memory for latent-condition inference, protocol-aware ephemeris reasoning for proactive mobility decisions, and satellite congestion prices for low-overhead inter-agent coordination without exchanging UE-level states or actions.
- We integrate mobility reliability through a constrained learning formulation in which a projected reliability multiplier regulates outage pressure during training, while retaining decentralized execution and the same protocol action space.
- We evaluate the framework across independently trained policies, multiple stochastic evaluation realizations, conventional mobility and duplication heuristics, and a 3GPP-inspired A3/TTT-based CHO baseline. We further consider overlapping traffic, capacity, and blockage disturbances to examine zero-shot behavior outside the nominal operating distribution, together with capability-level ablations that expose the mechanisms underlying memory and coordination.

II. Related Work

A. LEO Mobility, Conditional Handover, and Multi-Connectivity

Mobility management has long been recognized as a fundamental challenge of LEO satellite access networks because the serving opportunities of individual satellites are inherently time limited. Early approaches formulated satellite handover through graph- or flow-based optimization, using link quality, visibility duration, load, or service requirements to select subsequent serving satellites [5], [7]. Location-assisted and conditional-handover approaches further exploit the predictable motion of LEO satellites to trigger mobility procedures before the serving link becomes unusable [6]. AcceHo similarly exploits satellite mobility information to accelerate mobile-satellite handover [8], while SkyCastle addresses mobility-induced service disruption from a broader system perspective [9].

Multi-connectivity provides another route to service continuity by allowing multiple access links to coexist rather than relying exclusively on a single serving connection [16], [17]. Recent work such as GEMS explicitly incorporates the temporal evolution of satellite coverage into dual-connectivity configuration and balances instantaneous service improvement against future continuity [10]. These approaches highlight the value of retaining multiple candidate or serving links across mobility transitions.

At the protocol level, NR conditional reconfiguration and PDCP duplication provide complementary mechanisms for realizing such redundancy. CHO permits a target to be prepared before execution [24], while PDCP supports activation and deactivation of duplicated transmissions and duplicate-PDU handling [25]. However, standards define the protocol mechanisms rather than prescribing a LEO-specific policy for deciding when a UE should prepare a candidate, activate duplication, execute the switch, or release the reserved target under multi-user contention. Our work focuses precisely on this decision layer.

B. Learning-Based Mobility Control in LEO Networks

The rapid dynamics and large decision spaces of emerging mega-constellations have motivated learning-based mobility control. Deep reinforcement learning has been used to replace repeated online optimization with policies that can react rapidly after offline training. Liu *et al.* considered dynamic propagation conditions and limited satellite capacity and developed centralized and distributed multi-agent handover schemes [11]. Yang *et al.* formulated LEO handover as a decentralized decision problem and introduced multi-agent fingerprints with DDQN to mitigate learning non-stationarity and satellite load imbalance [12]. Other approaches employ DQN-based handover selection [13], safe learning for online NTN service provisioning [14], or hierarchical decision making that separates handover timing from target selection [15].

These approaches demonstrate that learning can effectively handle dynamic channel and load conditions that are difficult to capture with fixed thresholds. Nevertheless, the learned action is commonly the handover decision itself—for example, whether or where to switch. This leaves two issues particularly relevant to conditional mobility. First, preparation, duplication, execution, release, and rollback form a stateful protocol sequence whose consequences extend across several control epochs. Second, independently learned UE decisions interact through finite preparation slots and shared satellite capacity. The present work therefore treats mobility as a protocol-level multi-agent control problem rather than a sequence of isolated serving-satellite selections.

C. Agentic AI for Wireless and Satellite Networks

Agentic AI has recently emerged as a framework for moving wireless network intelligence beyond one-shot prediction or reactive optimization. Recent surveys describe agentic systems as autonomous decision makers that combine perception, reasoning, action, memory, and goal-directed interaction, and

identify advanced DRL as an important realization path for sequential wireless control [4], [18], [19].

Two broad implementation directions are emerging. One direction uses LLMs as semantic or strategic reasoning engines. The Think–Do architecture separates high-level LLM reasoning from real-time satellite resource execution [3], while agentic SAGIN frameworks employ LLM agents to interpret intents and coordinate specialized RL modules [20]. LLM-enhanced MAPPO has also been investigated for resource allocation in cognitive satellite–terrestrial networks [21]. Such architectures are particularly attractive when human intent, semantic knowledge, or cross-domain orchestration must be interpreted.

A complementary direction implements agentic capabilities directly through distributed learning and domain-specific decision modules. AERS models satellite beams as autonomous agents and combines recurrent multi-agent reinforcement learning with decentralized execution for beam hopping, user scheduling, and power allocation [2]. Agentic decision making has also been applied to intent-aware multipath routing and congestion control in large-scale LEO networks [22], and to task-oriented resource orchestration in satellite-assisted vehicular networks [23].

Our work belongs to the latter direction but addresses a different control layer. We do not employ an LLM to generate mobility actions at runtime. Instead, the agentic capabilities are instantiated by lightweight mechanisms matched to the structure of LEO mobility: recurrent local memory captures temporal experience, deterministic ephemeris provides a physics-grounded reasoning tool, and a scalar congestion price conveys shared-resource pressure. The resulting architecture therefore applies agentic control directly to the CHO/PDCP state machine at the control-epoch timescale, rather than to semantic orchestration, routing, or generic radio-resource scheduling.

III. System Model

We consider the downlink of a non-terrestrial network in which a set $\mathcal{L} = \{1, \dots, L\}$ of low Earth orbit (LEO) satellites serves a set $\mathcal{N} = \{1, \dots, N\}$ of ground user equipments (UEs). Time is slotted into control epochs $t \in \{0, 1, \dots, T-1\}$ of duration Δ; mobility decisions are taken once per epoch. Each UE is an autonomous decision-making agent that controls its own conditional handover (CHO) and packet data convergence protocol (PDCP) duplication procedure, while all UEs share the finite radio and control-plane resources of the satellites. Table I summarizes the parameters.

Modeling scope and protocol abstraction. Our objective is to study the decision coupling among target preparation, PDCP duplication, and primary-cell switching rather than to reproduce the complete signalling stack. Accordingly, the CHO procedure is represented by a system-level state machine that abstracts the preparation, execution, completion, and cooldown phases of conditional mobility, while PDCP duplication is represented by the synchronized transmission of the same packet stream over the serving and the prepared target legs. This abstraction is inspired by the NR conditional-reconfiguration procedure of 3GPP TS 38.331 and the PDCP functionality of 3GPP TS 38.323 [24], [25]. It does not model individual RRC messages, RLC entities, HARQ processes, or radio-link-failure recovery. The radio scheduler is kept fixed throughout all experiments, so that the learned agents control only mobility preparation, duplication, and switching.

A. Constellation Geometry and Visibility

Satellite ℓ moves on a circular orbit of altitude h and inclination ι. Its Earth-centred position is

$$\mathbf{p}_\ell(t) = (R_E + h)\big[\cos\theta_\ell,\ \sin\theta_\ell\cos\iota,\ \sin\theta_\ell\sin\iota\big]^{\mathsf{T}}, \quad (1)$$

with orbital phase $\theta_\ell(t) = \theta_\ell^0 + s_\ell\,\omega\,t\Delta$, angular rate $\omega = 2\pi/T_{\mathrm{orb}}$, initial phases θ_ℓ^0 evenly spaced over $[-0.38, 0.38]$ rad, and direction $s_\ell = +1$ for $\theta_\ell^0 \le 0$ and $s_\ell = -1$ otherwise. This construction represents a stylized local constellation slice with overlapping satellite passes rather than a full Walker constellation. It creates repeated rise–set transitions and simultaneous handover opportunities while keeping the mobility-control experiment computationally tractable. UE i is fixed at $\mathbf{u}_i$ on the Earth surface, with latitude and longitude drawn uniformly within a ± 0.04 rad $\times$ ± 0.05 rad area. The elevation angle $\epsilon_{i\ell}(t)$ and slant range $d_{i\ell}(t)$ follow from $\mathbf{p}_\ell(t) - \mathbf{u}_i$, and satellite ℓ is visible to UE i if $v_{i\ell}(t) = \mathbb{1}\{\epsilon_{i\ell}(t) \ge \epsilon_{\min}\} = 1$.

B. Channel and Link Abstraction

The received power of the link (i, ℓ) in dBm is

$$P_{i\ell}^{\mathrm{rx}}(t) = P^{\mathrm{tx}} + G^{\mathrm{tx}} + G^{\mathrm{rx}} - \mathrm{FSPL}\big(d_{i\ell}(t)\big) - L_{\mathrm{atm}} - L_{\mathrm{impl}} - L_{\mathrm{el}}\big(\epsilon_{i\ell}(t)\big) - \chi_{i\ell}(t) - B_{i\ell}(t) - Z_{i\ell}(t), \quad (2)$$

where $\mathrm{FSPL}(d) = 20\log_{10}(4\pi d f_c/c)$, $L_{\mathrm{el}}(\epsilon) = 1.5/\max\{0.2, \sin\epsilon\}$ dB models the additional low-elevation loss, and $\chi_{i\ell}(t)$ is log-normal shadowing with first-order autoregressive (AR(1)) temporal correlation and $w_{i\ell}(t) \sim \mathcal{N}(0,1)$,

$$\chi_{i\ell}(t) = \rho_\chi\,\chi_{i\ell}(t-1) + \sigma_\chi\sqrt{1-\rho_\chi^2}\,w_{i\ell}(t), \quad (3)$$

$B_{i\ell}(t)$ is the extra loss caused by blockage events and $Z_{i\ell}(t)$ is the loss of a hidden link-degradation regime (Section III-F). Let $P_N = -174 + 10\log_{10} W + F_{\mathrm{dB}}$ denote the receiver noise power in dBm, where F_{dB} is the receiver noise figure. Converting the received signal, noise, and aggregate interference powers to mW, the signal-to-interference-plus-noise ratio (SINR) of a visible link is

$$\gamma_{i\ell}(t) = \frac{10^{P_{i\ell}^{\mathrm{rx}}(t)/10}}{10^{P_N/10} + 10^{I_{\mathrm{dBm}}/10}}, \quad (4)$$

and it is set to a negligibly small value otherwise. Here I_{dBm} is a fixed aggregate interference floor; explicit co-channel interference dynamics are therefore outside the scope of the present model. The link is abstracted by a capped Shannon efficiency and a smooth block-error model,

$$\eta_{i\ell}(t) = \min\big\{\eta_{\max},\ 0.75\log_2\big(1+\gamma_{i\ell}(t)\big)\big\}, \quad (5)$$

$$q_{i\ell}(t) = \Big(1 + \exp\big(-0.75\,(\gamma_{i\ell}^{\mathrm{dB}}(t) + 2)\big)\Big)^{-1} v_{i\ell}(t), \quad (6)$$

where $q_{i\ell}$ is the per-packet delivery probability (50% at -2 dB).

C. Traffic, Scheduling, and Service

Packets of S bits arrive at UE i according to a Poisson process of rate $\lambda_i \kappa_i(t)$, where $\lambda_i = \bar{\lambda} u_i$ with $u_i \sim \mathcal{U}[1-\delta_\lambda, 1+\delta_\lambda]$ captures traffic heterogeneity and $\kappa_i(t) \geq 1$ is a traffic-burst multiplier. Packets wait in a finite queue $Q_i(t) \leq Q_{\max}$ and are dropped when their age exceeds the deadline D. A UE is active if $Q_i(t) > 0$, and its required rate is

$$R_i(t) = \min\big\{\lambda_i \kappa_i(t),\ Q_i(t) S/\Delta\big\}. \tag{7}$$

Each UE has a serving (primary) satellite $p_i(t)$ and, during a mobility procedure, a target (secondary) satellite $c_i(t)$. A UE requests a data leg on $p_i(t)$ and, when duplicating, a second leg on $c_i(t)$. The satellites run a fixed QoS-aware weighted-fair scheduler that is *not* controlled by the agents. For a leg (i,ℓ) the requested resource fraction is

$$f_{i\ell}(t) = \min\Big\{1,\ \frac{R_i(t)}{W\,\eta_{i\ell}(t)\,a_\ell(t)}\Big\}, \tag{8}$$

where $a_\ell(t) = \max\{0.05,\ a^0\, m_\ell(t)\}$ is the effective capacity scale and $m_\ell(t) \leq 1$ a capacity-drop multiplier. Let $n_\ell^{\mathrm{res}}(t)$ be the number of UEs holding a preparation reservation on ℓ; each reservation blocks a fraction ρ_{prep} of the satellite resource, leaving $A_\ell(t) = 1 - \rho_{\mathrm{prep}}\, n_\ell^{\mathrm{res}}(t)$ for data. With weights $w_{i\ell} = f_{i\ell}\,\varpi_i$ (ϖ_i: QoS priority), the granted share is $x_{i\ell} = w_{i\ell}$ if $\sum_j w_{j\ell} \leq A_\ell$ and $x_{i\ell} = A_\ell w_{i\ell} / \sum_j w_{j\ell}$ otherwise, which gives the leg rate $r_{i\ell}(t) = x_{i\ell}(t)\, W \eta_{i\ell}(t) a_\ell(t)$. The requested pressure of satellite ℓ is $\Phi_\ell(t) = \rho_{\mathrm{prep}} n_\ell^{\mathrm{res}}(t) + \sum_i f_{i\ell}(t)$. The scheduler is intentionally kept outside the learned action space. Although the formulation allows heterogeneous priority weights ϖ_i, all UEs use $\varpi_i = 1$ in the reported experiments; the evaluated scheduler is thus a demand-weighted fair allocator rather than a learned or heterogeneous-QoS scheduler. This separation allows performance differences among M1–M5 to be attributed to mobility preparation, duplication, reasoning, and coordination decisions rather than to changes in radio-resource scheduling.

Delivery. Without duplication, UE i transmits at $r_i = r_{ip_i}$ with success probability $q_i = q_{ip_i}$. With PDCP duplication, the same packets are sent on both legs, so that $r_i = \min\{r_{ip_i}, r_{ic_i}\}$ and

$$q_i^{\mathrm{dup}} = (1-\varkappa)\big(1 - \bar{q}_{ip_i}\bar{q}_{ic_i}\big) + \varkappa \max\{q_{ip_i}, q_{ic_i}\}, \tag{9}$$

where $\bar{q} = 1 - q$ and $\varkappa$ is a common-mode failure correlation. The number of delivered packets is $\mathrm{Bin}\big(\min\{\lfloor r_i \Delta/S \rfloor, Q_i\},\ q_i\big)$. We define the expected goodput $g_i = \min\{r_i, Q_i S/\Delta\}\, q_i$ and the realized goodput $\hat{g}_i$ (delivered bits per epoch).

Service metrics. The normalized service deficit and the outage indicator of an active UE are

$$\delta_i(t) = \frac{[R_i(t) - g_i(t)]^+}{R_i(t)}, \qquad o_i(t) = \mathbb{1}\big\{\hat{g}_i(t) < \beta\, R_i(t)\big\}, \tag{10}$$

where β is the minimum acceptable service ratio. Note that δ_i is a smooth expected-value quantity used for learning, whereas o_i is the realized, packet-level reliability event.

[TODO: Fig. 1: CHO/PDCP state machine (S, P, D, W, C) with actions on the edges, timers $T_{\mathrm{prep}}, T_{\mathrm{sw}}, T_{\mathrm{cd}}$, dwell guard, admission, and success probability (8).]

Fig. 1. Per-UE CHO and PDCP-duplication state machine.

D. Conditional Handover and PDCP Duplication Protocol

Each UE runs the five-state mobility state machine of Fig. 1: SINGLE (S), PREPARED (P), DUPLICATING (D), SWITCHING (W), and COOLDOWN (C). The set of candidate targets $\mathcal{C}_i(t)$ contains the K visible non-serving satellites with the highest *observed* SINR (Section III-E). The per-UE action set is

$$\mathcal{A} = \{\textsc{Stay},\ \textsc{Prep}_1, \ldots, \textsc{Prep}_K,\ \textsc{DupOn},\ \textsc{DupOff},\ \textsc{Commit},\ \textsc{Release},\ \textsc{Rollback}\}, \tag{11}$$

so $|\mathcal{A}| = K + 6$. A state-dependent hard mask $\mathbf{m}_i(t) \in \{0,1\}^{|\mathcal{A}|}$ enforces protocol and physical validity:

- In S, $\textsc{Prep}_k$ requests CHO preparation towards the k-th candidate. If admitted, the UE enters P with a preparation timer T_{prep}.
- In P (after the timer expires), DUPON activates duplication (P→D), COMMIT executes the CHO (P→W), and RELEASE cancels the preparation (P→S).
- In D, DUPOFF returns to P, COMMIT executes the CHO, and ROLLBACK cancels both the duplication and the preparation.
- COMMIT is feasible only if the target is visible and the dwell condition $t - t_i^{\mathrm{sw}} \geq T_{\mathrm{dwell}}$ holds, where t_i^{sw} is the epoch of the last completed switch.

If the target becomes invisible while the UE is in P or D, the preparation is lost and the UE returns to S. After T_{sw} epochs in W, the switch succeeds with probability

$$p_i^{\mathrm{HO}} = p_0\, q_{ic_i}(t)\, \exp\big(-[\Phi_{c_i}(t) - 1]^+\big), \tag{12}$$

which degrades with poor target quality and target congestion. On success, c_i becomes the new serving satellite and the UE enters C for T_{cd} epochs; otherwise it returns to S on the old serving satellite. A successful switch back to the previous serving satellite within T_{pp} epochs is counted as a ping-pong.

Preparation admission. CHO preparation consumes two separate resources of the target satellite: a data-plane reservation ρ_{prep} (above) and one of M control-plane preparation slots. All preparation requests of an epoch are collected first; if the number of new requests towards satellite ℓ exceeds $\min\{M - n_\ell^{\mathrm{res}},\ \lfloor A_\ell/\rho_{\mathrm{prep}} \rfloor\}$, the admitted requests are drawn uniformly at random and the others are rejected. Admission is therefore an explicit multi-UE contention.

E. Information Structure and Partial Observability

The agents do not observe the physical state. Actor telemetry is delayed by τ_{obs} epochs and corrupted by temporally correlated noise:

$$\tilde{\gamma}_{i\ell}^{\mathrm{dB}}(t) = \gamma_{i\ell}^{\mathrm{dB}}(t-\tau_{\mathrm{obs}}) + n_{i\ell}(t), \qquad (13)$$
$$n_{i\ell}(t) = \rho_n n_{i\ell}(t-1) + \sigma_\gamma \sqrt{1-\rho_n^2}\, w'_{i\ell}(t),$$

and analogously for the observed spectral efficiency $\tilde{\eta}_{i\ell}$ and the observed load $\tilde{\Phi}_{i\ell} = [\Phi_\ell(t-1-\tau_{\mathrm{obs}}) + n_{i\ell}^{\Phi}(t)]^+$. The local observation $\mathbf{o}_i(t) \in \mathbb{R}^{11+5+7K}$ contains (i) the queue length, head-of-line delay, arrival rate, reliability-class indicator, duplication flag, one-hot protocol state, and protocol timer; (ii) for the serving satellite: visibility, $\tilde{\gamma}$, $\tilde{\eta}$, elevation, and $\tilde{\Phi}$; and (iii) for each candidate: presence, a prepared-target flag, $\tilde{\gamma}$, $\tilde{\eta}$, elevation, $\tilde{\Phi}$, and the observed SINR trend $\tilde{\gamma}(t)-\tilde{\gamma}(t-1)$. Agents do *not* observe the remaining visibility time, the capacity multipliers m_ℓ, the event realizations, the latent link regimes, or the states and actions of other UEs. The candidate list is ranked by observed current SINR only, and the action mask does not check whether a candidate will remain visible long enough to complete a handover: anticipating the satellite geometry is left to the decision layer. The per-UE reliability-class indicator is fixed to 0.99 in the present experiments and should not be confused with the system-level outage constraint $\varepsilon_{\mathrm{out}} = 0.10$ introduced in Section IV.

Per-epoch control sequence. At decision epoch t, UE i forms its local observation from delayed telemetry, its current protocol state, and its locally experienced history; in M4 and M5 it additionally receives the satellite congestion prices computed at the end of the preceding epoch. Each UE then selects one masked mobility action independently. The environment first resolves ongoing protocol timers and switching outcomes, re-validates the selected actions (actions that have become infeasible are replaced by STAY), and jointly admits the simultaneous preparation requests. New packets arrive, the fixed scheduler allocates the remaining satellite resources to the serving and duplicated data legs, and packet delivery, service deficit, and outage are realized. The resulting local outcomes are fed to the recurrent memory for the next decision. Finally, the hidden regimes and disturbance events advance, the links and telemetry are refreshed, and the satellite coordinator updates its congestion price from the aggregate intended demand, so that the new price affects only the next decentralized decision. Coordination is therefore causal and one-epoch delayed rather than based on instantaneous access to the actions of other UEs.

F. Non-Stationarity: Hidden Regimes and Disturbance Events

Hidden link regimes. Every UE–satellite path (i,ℓ) follows a hidden semi-Markov regime GOOD $\to$ DEGRADED $\to$ RECOVERING $\to$ GOOD. A path enters DEGRADED with probability p_{deg} per epoch (and with probability p_{deg}^0 at the start of an episode), stays there for $U\{8,\dots,24\}$ epochs with an extra loss $Z_{i\ell} \sim \mathcal{U}[7,15]$ dB, and then recovers over $U\{4,\dots,10\}$ epochs during which $Z_{i\ell}$ decays linearly to zero. These regimes persist over many epochs and are observable only through their delayed, noisy effect on $\tilde{\gamma}$, which is what makes memory useful.

TABLE I
SIMULATION AND LEARNING PARAMETERS

Network and protocol	
Satellites / UEs / candidates	$L=6$, $N=12$, $K=3$
Altitude, inclination, period	600 km, 53°, $T_{\mathrm{orb}} = 5800$ s
Epoch, episode, min. elev.	$\Delta = 1$ s, $T = 420$, $\epsilon_{\min} = 10°$
Carrier, bandwidth	$f_c = 2$ GHz, $W = 4$ MHz
Tx power, Tx/Rx gain	46 dBm, 30/0 dBi
Noise figure, interference	$F_{\mathrm{dB}} = 7$ dB, $I_{\mathrm{dBm}} = -108$
Losses (atm./impl.)	0.5/3 dB
Shadowing	$\sigma_\chi = 2$ dB, $\rho_\chi = 0.95$
SE cap, capacity scale	$\eta_{\max} = 6$, $a^0 = 0.7$
Packet size, deadline, queue	$S = 12$ kbit, $D = 2$ s, $Q_{\max} = 400$
Mean rate, spread	$\bar{\lambda} = 2$ Mb/s, $\delta_\lambda = 0.35$
Outage ratio, reliab. class	$\beta = 0.8$, 0.99 (all UEs)
$T_{\mathrm{prep}}, T_{\mathrm{sw}}, T_{\mathrm{cd}}, T_{\mathrm{dwell}}, T_{\mathrm{pp}}$	1, 2, 5, 15, 20 epochs
Prep. reservation, control slots	$\rho_{\mathrm{prep}} = 0.01$, $M = 2$
HO success, dup. corr.	$p_0 = 0.995$, $\varkappa = 0.15$
Partial observability and non-stationarity	
Telemetry lag, noise	$\tau_{\mathrm{obs}} = 2$, $\sigma_\gamma = 1.5$ dB, $\sigma_\Phi = 0.08$, $\rho_n = 0.8$
Hidden regime	$p_{\mathrm{deg}}^0 = 0.06$, $p_{\mathrm{deg}} = 0.006$
Blockage (2 events)	8–15 dB, 25–55 epochs, 30–65% of UEs
Capacity drop (2 events)	$m = 0.45$–0.80, 25–70 epochs
Traffic burst (1 event)	$\kappa = 1.35$–1.85, 35–70 epochs
Objective and agentic modules	
Cost weights	$\lambda = 0.25$, $\eta_{\mathrm{prep}} = 0.2$, $\eta_{\mathrm{HO}} = 0.05$
Outage target	$\varepsilon_{\mathrm{out}} = 0.10$
Dual step, bound	$\alpha_\mu = 0.005$, $\mu_{\max} = 10$
Price step, $y_{\max}$	$\alpha_y = 0.05$, 5
Reasoning horizon, discount	$H = 6$, $\xi = 0.95$
Learning (CTDE-MAPPO)	
Actor / critic MLP	$[128,128]$ / $[256,256]$
GRU hidden size, TBPTT length	128, 16
Rollout, iterations	$T_{\mathrm{roll}} = 128$, $I = 2000$
PPO epochs, minibatch	5, 128
γ, λ_{GAE}, clip, KL	0.99, 0.95, 0.2, 0.03
LR actor/critic	$3{\cdot}10^{-4} \to 3{\cdot}10^{-5}$ / $10^{-3} \to 10^{-4}$
Entropy / value coef., grad. norm	0.01 / 0.5, 0.5

Disturbance events. Each training and nominal-evaluation episode contains five events whose start, duration, severity, affected satellite, and affected UE subset are sampled from the ranges in Table I: two blockages (extra loss $B_{i\ell}$ on the selected UE–satellite links), two capacity drops (multiplier $m_\ell < 1$ on one satellite), and one traffic burst ($\kappa_i > 1$ on a subset of UEs). For the *compound-stress* evaluation, the randomized training-event schedule is replaced by a deterministic overlapping disturbance pattern: a traffic burst ($\kappa = 2.5$ on 50% of the UEs, $t \in [60,120)$), a capacity drop ($m_2 = 0.45$, $t \in [80,130)$), and an 18-dB blockage affecting all links of 25% of the UEs during $t \in [100,125)$. The traffic-burst and blockage severities lie outside their training ranges, and the blockage affects all satellites of a UE instead of a single one, whereas the capacity-drop magnitude equals the most severe value of its training range. More importantly, the three disturbances overlap in time, creating a compound operating condition not encountered during training.

IV. Problem Formulation

A. Per-Epoch Costs

Let $\mathcal{N}^{\mathrm{act}}(t)$ be the set of active UEs. We define the service, control, and reliability costs of epoch t as

$$C_{\mathrm{def}}(t) = \frac{1}{|\mathcal{N}^{\mathrm{act}}(t)|} \sum_{i \in \mathcal{N}^{\mathrm{act}}(t)} \delta_i(t), \tag{14}$$

$$C_{\mathrm{dup}}(t) = \frac{B^{\mathrm{dup}}(t)}{B^{\mathrm{use}}(t) + B^{\mathrm{dup}}(t)}, \tag{15}$$

$$C_{\mathrm{prep}}(t) = \frac{\rho_{\mathrm{prep}}}{L} \sum_{\ell \in \mathcal{L}} n_\ell^{\mathrm{res}}(t), \tag{16}$$

$$C_{\mathrm{HO}}(t) = \frac{1}{|\mathcal{N}^{\mathrm{act}}(t)|} \sum_{i \in \mathcal{N}^{\mathrm{act}}(t)} \mathbb{1}\{a_i(t) = \textsc{Commit}\}, \tag{17}$$

$$C_{\mathrm{ctrl}}(t) = C_{\mathrm{dup}}(t) + \eta_{\mathrm{prep}} C_{\mathrm{prep}}(t) + \eta_{\mathrm{HO}} C_{\mathrm{HO}}(t), \tag{18}$$

$$C_{\mathrm{out}}(t) = \frac{1}{|\mathcal{N}^{\mathrm{act}}(t)|} \sum_{i \in \mathcal{N}^{\mathrm{act}}(t)} o_i(t), \tag{19}$$

where $B^{\mathrm{use}}(t)$ denotes the volume of the original PDCP packet stream scheduled for transmission during epoch t, before packet-delivery errors are realized, and $B^{\mathrm{dup}}(t)$ denotes the additional transmission volume generated by sending a copy of that stream over the secondary leg. Hence, C_{dup} measures transmission redundancy rather than successfully delivered traffic. C_{ctrl} captures the resource and signalling overhead of mobility control: duplicated transmissions, reserved-but-unused target resources, and handover executions.

B. Constrained Decentralized POMDP

The problem is a cooperative, constrained decentralized partially observable Markov decision process (Dec-POMDP) $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}\}, \mathcal{P}, r, \{\Omega\}, \mathcal{O}, \gamma \rangle$. The state $s(t) \in \mathcal{S}$ comprises the satellite geometry, channel and shadowing states, hidden regimes, event realizations, queues, and protocol states of all UEs; $\mathcal{P}$ is given by Section III; and each agent receives the local observation $\mathbf{o}_i(t) \sim \mathcal{O}(\cdot \mid s(t))$ of Section III-E together with its action mask. All agents share the team reward

$$r(t) = -C_{\mathrm{def}}(t) - \lambda\, C_{\mathrm{ctrl}}(t). \tag{20}$$

Reliability is imposed as a constraint on the long-run outage ratio rather than as a further reward weight:

$$\max_{\boldsymbol{\pi}} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \gamma^t r(t)\Big] \ \text{ s.t. } \ \mathbb{E}_{\boldsymbol{\pi}}\big[C_{\mathrm{out}}(t)\big] \le \varepsilon_{\mathrm{out}}, \tag{21}$$

where $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_N)$ are decentralized policies $\pi_i(a_i \mid \mathbf{o}_i^{\le t})$ that may depend on the local observation history. The per-satellite capacity constraints are enforced by the scheduler and the admission rule and need not be learned. Problem (21) is solved through its Lagrangian relaxation

$$\max_{\boldsymbol{\pi}} \min_{\mu \ge 0} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \gamma^t r_\mu(t)\Big], \tag{22}$$

$$r_\mu(t) = r(t) - \mu\big(C_{\mathrm{out}}(t) - \varepsilon_{\mathrm{out}}\big).$$

Why the problem is hard. Three features make (21) difficult for conventional CHO rules and for memoryless learning: (i) *partial observability* with delayed, noisy telemetry and hidden, persistent link regimes; (ii) *delayed consequences*: a preparation blocks resources and control slots before any benefit appears, the dwell guard restricts future switches, and a target chosen now may set before the switch completes; and (iii) *coupling*: UEs compete for preparation slots, reservations, and data capacity, and congestion lowers the handover success probability (12).

TABLE II
Capability-structured model family

Model	Capability added	Actor input	Training reward
M1	Perception (core)	$\mathbf{o}_i$	r
M2	+ Memory	$[\mathbf{o}_i, \mathbf{h}_i]$	r
M3	+ Reasoning	$[\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i]$	r
M4	+ Coordination	$[\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i, \mathbf{y}_i]$	r
M5	+ Goal regulation	$[\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i, \mathbf{y}_i, \mu/\mu_{\max}]$	r_μ

V. Proposed Capability-Structured Agentic Framework

We operationalize agentic control through a structured set of capabilities rather than treating agentic AI as synonymous with an LLM-based controller. The architecture combines a learned decentralized policy with lightweight domain-specific mechanisms that exploit the structure of LEO mobility, following the four capabilities of agentic AI. *Perception* is provided by masked, telemetry-based local observation and per-UE autonomous action selection; *memory* by a recurrent state that infers persistent hidden conditions from delayed feedback; *reasoning* by a protocol-aware ephemeris lookahead that anticipates the delayed consequences of preparation and switching; and *coordination* by a low-dimensional satellite congestion price that summarizes shared-resource pressure without exchanging UE actions. Finally, a training-time reliability multiplier provides explicit *goal regulation*.

These capabilities are introduced incrementally to form the M1–M5 model family in Table II. The resulting ladder is an architectural decomposition used for capability-structured evaluation, not an assumption that performance must improve monotonically from M1 to M5. Each variant is trained independently in the same environment with the same base reward, except for the reliability Lagrangian used by M5. No language model is required, and online execution remains numerical and lightweight: actions are produced locally by each UE, while satellites broadcast only the scalar congestion prices required by M4 and M5.

A. Perception and Core Decision Making (M1)

All UEs share one stochastic actor π_θ and are trained with centralized training and decentralized execution (CTDE) using multi-agent proximal policy optimization (MAPPO) [26]. The actor maps its input $\mathbf{z}_i(t)$ to logits over $\mathcal{A}$, and invalid actions are removed by a masked softmax,

$$\pi_\theta(a \mid \mathbf{z}_i, \mathbf{m}_i) = \frac{m_{i,a} \exp\big(\ell_\theta(\mathbf{z}_i)_a\big)}{\sum_{a'} m_{i,a'} \exp\big(\ell_\theta(\mathbf{z}_i)_{a'}\big)}. \tag{23}$$

Perception in M1 is the masked, telemetry-based observation $\mathbf{z}_i = \mathbf{o}_i$. The centralized critic $V_\psi(\mathbf{s}^c)$ is used only during

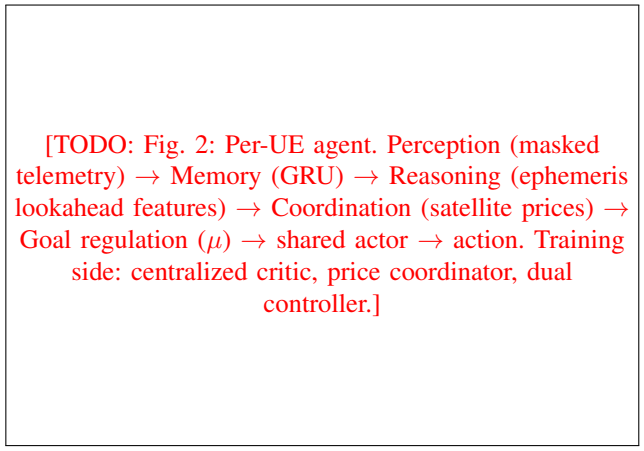


Fig. 2. Architecture of the capability-structured agent (execution: solid; training only: dashed).

training. Its input $\mathbf{s}^c(t)$ concatenates the observations of all UEs with the true per-satellite utilization and pressure Φ_ℓ, and, when available, the satellite prices and the dual variable.

B. Memory: Inferring Hidden Regimes (M2)

Because link regimes are hidden and telemetry is delayed, the current observation is not a sufficient statistic. M2 equips every agent with a gated recurrent unit (GRU) that summarizes its interaction history,

$$\mathbf{h}_i(t) = \mathrm{GRU}_\theta\big(\mathbf{h}_i(t-1),\ [\mathbf{o}_i(t),\ \mathbf{e}(a_i(t-1)),\ \boldsymbol{\zeta}_i(t-1)]\big), \tag{24}$$

where $\mathbf{e}(\cdot)$ is a one-hot encoding and $\boldsymbol{\zeta}_i \in \mathbb{R}^6$ is the agent's own *experienced outcome*: the last handover result ($+1$ success, -1 failure, 0 none), its outage indicator o_i, its deficit δ_i, the duplication gain $[g_i - g_i^{\mathrm{prim}}]^+/R_i$, its resource-usage indicator, and the experienced congestion $\Phi/(1+\Phi)$ of its serving or target satellite. The actor input becomes $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{h}_i]$. Memory thus supports long-horizon adaptation to non-stationary link conditions using only locally available feedback. The GRU is trained end-to-end with truncated backpropagation through time.

C. Reasoning: Protocol-Aware Ephemeris Lookahead (M3)

The mask does not prevent choosing a target that is about to set, and the benefit of a preparation appears only several epochs later. M3 therefore adds an analytical planner that exploits the one piece of information that is deterministic in LEO systems: the satellite ephemeris. For every feasible first action a, the planner evaluates a small library $\Pi_i(a)$ of protocol-consistent plans of length H (e.g., PREP$_k$–wait–COMMIT; PREP$_k$–wait–DUPON–COMMIT; STAY), padded with STAY, by rolling the UE's own protocol state machine forward. The future SINR of link (i,ℓ) is extrapolated from the *observed* SINR and the ephemeris-predicted range,

$$\hat{\gamma}_{i\ell}^{\mathrm{dB}}(t+\tau) = \tilde{\gamma}_{i\ell}^{\mathrm{dB}}(t) - 20\log_{10}\frac{d_{i\ell}(t+\tau)}{d_{i\ell}(t)}, \tag{25}$$

and set to zero once $\epsilon_{i\ell}(t+\tau) < \epsilon_{\min}$. The predicted leg rate uses the nominal capacity a^0 scaled by the observed load $\tilde{\Phi}_{i\ell}$, from which the planner obtains the predicted deficit $\hat{\delta}_i(t+\tau)$, including the duplication diversity of (9). The cost of a first action is the best discounted plan cost,

$$J_i(a) = \min_{\varsigma\in\Pi_i(a)} \sum_{\tau=0}^{H-1} \xi^\tau \Big(\hat{\delta}_i(t+\tau \mid \varsigma) + \mathbb{1}_{\mathrm{coord}}\, \mathbf{y}(t)^\top \big[\hat{\mathbf{u}}_i(t+\tau \mid \varsigma) - \hat{\mathbf{u}}_i^{\mathrm{STAY}}(t+\tau)\big]^+\Big), \tag{26}$$

where $\mathbb{1}_{\mathrm{coord}} = 1$ for M4 and M5 (and 0 otherwise), and the second term prices the *incremental* per-satellite resource demand $\hat{\mathbf{u}}_i$ of the plan relative to staying. The reasoner deliberately ranks plans by predicted service continuity rather than reproducing the complete training reward: in M3 the second term of (26) is absent, so resource and signalling overheads are learned through the MAPPO team reward rather than hard-coded into the planner, and M4 adds only the marginal congestion-price term, which lets the same geometry-aware reasoner account for the external resource pressure created by multi-UE decisions.

Let $J_i^{\min}$ and $J_i^{\max}$ be the minimum and maximum of $J_i(a)$ over the feasible actions. The reasoning feature is

$$\phi_{i,a} = \begin{cases} \dfrac{J_i(a) - J_i^{\min}}{J_i^{\max} - J_i^{\min} + \epsilon_\phi}, & J_i^{\max} - J_i^{\min} > \epsilon_\phi, \\ 0, & \text{otherwise}, \end{cases} \tag{27}$$

for feasible actions, clipped to $[0,1]$, and $\phi_{i,a} = 1$ for masked actions. The vector $\boldsymbol{\phi}_i \in [0,1]^{|\mathcal{A}|}$ is appended to the actor input: $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i]$. The planner is *advisory*: the actor learns how much to trust it. It has no access to future events, hidden regimes, true capacity multipliers, or the states of other UEs.

D. Coordination: Congestion Prices (M4)

UEs compete for preparation slots, reservations, and data capacity, but exchanging intentions between UEs is impractical. M4 coordinates them through a lightweight price signal maintained by each satellite. Satellite ℓ measures its aggregate intended demand

$$U_\ell(t) = \sum_i f_{i\ell}^{\mathrm{serv}}(t) + \frac{n_\ell^{\mathrm{req}}(t)}{M} + \sum_i f_{i\ell}^{\mathrm{dup}}(t), \tag{28}$$

i.e., the serving demand, the preparation requests normalized by the number of control slots, and the duplication demand, and updates its price by projected subgradient ascent,

$$y_\ell(t+1) = \Big[y_\ell(t) + \alpha_y\big(U_\ell(t) - 1\big)\Big]_0^{y_{\max}}. \tag{29}$$

Prices rise while demand exceeds capacity and decay otherwise; they are reset at the start of each episode and are not learned. Each UE receives the normalized prices of its serving satellite and its K candidates, $\mathbf{y}_i = [y_{p_i}, y_{c_{i1}}, \ldots, y_{c_{iK}}]/y_{\max}$, which enter both the actor input and the reasoning cost (26). The coordinator never allocates actions: decisions remain fully decentralized, and the price is observed with a one-epoch lag. The signalling cost is one scalar per satellite and epoch. The price should therefore be interpreted as a compressed coordination context rather than a centralized

Algorithm 1 Training of the capability-structured agents

Initialize θ (actor, GRU), ψ (critic), $\mu \leftarrow 0$
for iteration $= 1, \ldots, I$ **do**
  for $t = 1, \ldots, T_{\text{roll}}$ **do**
    Each UE forms $\mathbf{o}_i(t)$ and mask $\mathbf{m}_i(t)$ ▷ perception
    $\mathbf{h}_i(t) \leftarrow \text{GRU}_\theta(\cdot)$ ▷ M2+
    $\boldsymbol{\phi}_i(t) \leftarrow$ lookahead (26) with prices $\mathbf{y}(t)$ ▷ M3+
    $a_i(t) \sim \pi_\theta(\cdot \mid \mathbf{z}_i(t), \mathbf{m}_i(t))$ for all i in parallel
    Environment step; observe $r(t)$, $C_{\text{out}}(t)$, outcomes $\boldsymbol{\zeta}_i(t)$
    Satellites update prices by (29) ▷ M4+
  end for
  Compute r_μ with the current μ ▷ M5
  Update θ, ψ with PPO, GAE, and TBPTT
  Update μ by (30) ▷ M5
end for

allocation command: it does not reveal individual UE states, actions, queues, or future events, and communicates only whether the aggregate intended demand has recently exceeded the normalized satellite resource level.

E. Goal Regulation: Primal–Dual Reliability Control (M5)

M5 addresses the Lagrangian problem (22). After every rollout of T_{roll} epochs, the multiplier is updated by the following projected multiplier step on the realized outage ratio of the rollout,

$$\mu \leftarrow \Big[\mu + \alpha_\mu \big(\bar{C}_{\text{out}}^{\text{roll}} - \varepsilon_{\text{out}}\big)\Big]_0^{\mu_{\max}}, \tag{30}$$

and the policy is trained on r_μ. The multiplier is held fixed within each rollout, updated between policy iterations, and frozen after training. In our implementation, μ does not modify the ranking of the analytical lookahead; reliability enters the learned decision through the Lagrangian reward, the recurrently observed outcomes, and the normalized feature $\mu/\mu_{\max}$ supplied to the actor and the critic. Consequently, $\mu = 0$ makes the *instantaneous training reward* identical to the M4 base reward, but a trained M5 network remains a separately optimized policy and is not identical to M4.

F. Training Procedure and Complexity

Algorithm 1 summarizes training. The shared actor and the centralized critic are optimized with the clipped PPO objective [27] and generalized advantage estimation [28] on the team reward (r or r_μ), with early stopping at a target KL divergence and linearly annealed learning rates.

Execution complexity. In each control epoch, a UE performs one recurrent-state update and one actor forward pass. For M3–M5, the analytical reasoner evaluates at most $\max\{2K+1, 4\}$ protocol-consistent plans, each over a horizon of H epochs. Its per-agent planning complexity is therefore $\mathcal{O}(KH)$ and does not grow with the total number of UEs. Across all N agents, the aggregate reasoning work is $\mathcal{O}(NKH)$, but the N per-UE computations are mutually independent and can be executed in parallel. Coordination requires only one congestion price per satellite and epoch, and the M5 reliability multiplier introduces no execution-time inter-UE signalling. [TODO: After latency_bench: add "Measured on a single CPU thread, the decision latency per epoch for all $N = 12$ UEs is … ms, well below $\Delta = 1$ s."]

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